A Macroeconomic Model of the Cross-section of Currencies

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motivation

What drives the cross sectional variation of bilateral exchange rates?

International finance view: Verdelhan (2018), Lustig and Richmond (2020), ...

- two factors account for most of variation
- countries heterogeneously exposed to factors

International macro advances: Gabaix Maggiori (2015), Itskhoki Mukhin (2021),...

- models that explain exchange rate variations and connection to macro
- focus on single exchange rate, typically dollar

This paper:

- structural macro model that explains factor structure of exchange rates & their drivers

what we do

Macro model of the cross-section of currencies:

- aggregate shocks to output and risk-taking capacity
- heterogeneous countries: dollar assets, exposure to commodity cycle
- exchange rates have factor structure: "dollar" and "commodity" factors

Empirical analysis:

- dollar and commodity factors explain around 50% of variation in ΔXR
- factor loadings explained by dollar assets and export share of commodities

Quantitative analysis:

- factors reflect combination of shocks to global risk premium and economic activity
- dollar factor = US non-traded output + risk, commodity factor = global traded output + risk
- **xr drivers differ by country** depending on commodity exposure and dollar assets

related literature

- factor structure in exchange rates and cross-country heterogeneity:

  Lustig, Roussanov, and Verdelhan (2011), Hassan (2013), Lettau, Maggiori, and Weber (2014), Lustig, Roussanov, and Verdelhan (2014), Ready, Roussanov, and Ward (2017), Verdelhan (2018), Richmond (2019), Lustig and Richmond (2020), Koijen and Yogo (2020), ...

- exchange rates in macro:

  Backus and Smith (1993), Lustig and Verdelhan (2007), Colacito and Croce (2011), Menkhoff, Sarno, Schmeling, and Schrimpf (2012), Lustig and Verdelhan (2019), Chernov et al. (2024), ...

- exchange rates and financial frictions:

  Alvarez, Atkeson, and Kehoe (2002, 2009), Gabaix and Maggiori (2015), Schmitt-Grohé and Uribe (2017), Itskhoki and Mukhin (2021), Itskhoki and Mukhin (2025), Engel and Wu (2024), Chahrour, Cormun, De Leo, Guerrón-Quintana, and Valchev (2024), Kekre and Lenel (2025), ...

Model

households: preferences and endowments

Continuum of small open economies indexed by $i \in [0,1]$, household + central bank

$$\max \mathbb{E}_0 \sum_{t=0}^{t=\infty} \beta^t u(\mathcal{C}(C_{it}^N, C_{it}^T))$$

$$P_{it}^N C_{it}^N + C_{it}^T = P_{it}^N Y_{it}^N + Y_{it}^T + R_{it} Q_{it} B_{it} - Q_{it} B_{i,t+1} + T_{it} + \Pi_{it}$$

- P_{it}^N price of non-traded good, Q_{it} price of consumption basket $\mathcal{C}(C_{it}^N, C_{it}^T) = (C_{it}^N)^\alpha (C_{it}^T)^{1-\alpha}$
- save in local currency bond $B_{i,t+1}$ at R_{it}, get fiscal rebate T_{it}, profits of intermediaries Π_{it}
- non-traded endowment: $N_{it} = N(1 + x_{it})$, idiosyncratic shocks x_{it}
- tradable endowment $Y_{it}^T = 1 + e_i z_t$: country-specific exposure e_i to global shock z_t

tradable endowments and commodities

Microfoundation for $Y_{it}^T = 1 + e_i z_t$:

- two tradable goods: primary input (commodities) and final good
- final goods produced with primary inputs and labor
- primary inputs in heterogeneous inelastic supply
- shocks to productivity of primary inputs maps into z_t
- endowments of primary inputs map into e_i

central banks

Buy foreign reserves $M_{i,t+1}$ denominated in USD, transfers net proceedings to households

$$T_{it} = R_{ut} Q_{ut} M_{it} - Q_{ut} M_{i,t+1}$$

Country resource constraint:

$$Q_{it} C_{it} = P_{it}^N Y_{it}^N + Y_{it}^T + Q_{it}(R_{it} B_{i,t} - B_{i,t+1}) + Q_{ut}(R_{ut} M_{it} - M_{i,t+1}) + \Pi_{it}$$

Reserve policy (dollar bonds):

$$Q_{ut} M_{i,t+1} = \tau \cdot Q_t M_i + (1 - \tau) \cdot Q_{ut} M_i$$

- here $Q_t \equiv \int_0^1 Q_{it}$ is a diversified currency basket
- exogenous heterogeneity in reserve level M_i
- captures domestic policy uncertainty, see e.g. Drenik, Kirpalani, and Perez (2022)

global financial intermediaries

A zero-capital portfolio to invest in local currency in all countries, including the US

- pick local currency bonds $A_{i,t+1}$, their USD value denoted by $L_{i,t+1} \equiv Q_{it}/Q_{ut} \cdot A_{i,t+1}$
- choose portfolio of bonds $\{L_{i,t+1}\}$ in different currencies: maximize

$$\mathbb{E}_t\left[\Lambda_{t+1}\int L_{i,t+1}\underbrace{\left(\frac{R_{it}}{R_{ut}}\frac{Q_{i,t+1}}{Q_{u,t+1}}\frac{Q_{ut}}{Q_{it}}-1\right)}_{\text{excess return vs USD}}di\right]-\underbrace{\frac{1}{2\chi}\int\left(L_{i,t+1}-L_{t+1}\right)^2 di}_{\text{portfolio cost}}$$

- benchmark level L_{t+1} set at total reserve demand $\int M_{i,t+1}di$

intermediaries' stochastic discount factor

Intermediaries use a stochastic discount factor that combines tradable SDF of households:

$$\log \Lambda_{t+1} = (1-\Gamma_t)\lambda_t + \Gamma_t \left(\eta \int \log \Lambda^T_{i,t+1} di + (1-\eta) \log \Lambda^T_{u,t+1} \right)$$

Tradable SDF is captures marginal utility of tradables:

$$\Lambda^T_{i,t+1} \equiv \beta \frac{\partial u(C_{i,t+1})}{\partial C^T_{i,t+1}} \left(\frac{\partial u(C_{it})}{\partial C^T_{it}} \right)^{-1}$$

$\Gamma_t = \Gamma(1+\gamma_t)$ sets risk aversion, λ_t sets $\mathbb{E}_t[\Lambda_{t+1}] = \eta \int \mathbb{E}_t[\Lambda^T_{i,t+1}] di + (1-\eta)\mathbb{E}_t[\Lambda^T_{u,t+1}]$

equilibrium

Equilibrium is a set of processes $\{P_{it}^N, Q_{it}, R_{it}\}$ and $\{C_{it}^N, C_{it}^T, B_{i,t+1}, L_{i,t+1}, M_{i,t+1}\}$ such that the quantities solve the optimization problems and the following markets clear:

non-tradable goods i: $$C_{it}^N = Y_{it}^N$$

tradable goods: $$\int C_{it}^T di = \int Y_{it}^T di$$

currency i bonds: $$A_{i,t+1} + B_{i,t+1} = 0$$

dollar bonds: $$\int \underbrace{M_{i,t+1} di}_{\text{reserves}} \underbrace{+ L_{u,t+1}}_{\text{intermediary}} \underbrace{+ B_{u,t+1}}_{\text{US HH}} = 0$$

Shocks: $z_{t+1} = \rho_z z_t + \sigma_z \epsilon_{z,t+1}$, $x_{i,t+1} = \rho_x x_{it} + \sigma_x \epsilon_{i,t+1}$, and $\gamma_{t+1} = \rho_\gamma \gamma_t + \sigma_\gamma \epsilon_{\gamma,t+1}$

Solving the model

solution

Linear approximation around steady state in the limit of small shocks, large risk aversion

- simultaneously $(\sigma_z, \sigma_x, \sigma_\gamma) \longrightarrow 0$ and $\Gamma \longrightarrow \infty$ with $(\Gamma\sigma_z^2, \Gamma\sigma_x^2, \Gamma\sigma_\gamma^2) \longrightarrow (\Gamma_z, \Gamma_x, \Gamma_\gamma)$
- steady state symmetric in everything but inflows $\{L_i\}_i$ and L_u
- linear equilibrium of the form

$$q_{it} = \eta_i x_{it} + \underbrace{\theta_i z_t + \xi_i x_{ut} + \mu_i \gamma_t}_{\text{global exogenous shocks}} + \underbrace{\omega_i l_{it} + \delta_i l_{ut} + \zeta_i m_t}_{\text{endogenous states}}$$

$$q_{ut} = \theta_u z_t + \xi_u x_{ut} + \mu_u \gamma_t + \omega_u l_{ut} + \zeta_u m_t$$

- US non-traded endowment shock affects global finance through US exchange rate

steady-state positions

Proposition 1

Suppose $\eta = 1/2$ and $e_u = e$. In the steady state,

$$L_i - L = -\underbrace{\chi\Gamma\sigma_z^2 e(\alpha + (1-\alpha)\rho)}_{\text{exposure to } z_t} \cdot (\theta_i - \theta_u) - \underbrace{\frac{\chi\Gamma\sigma_x^2\alpha(\rho-1)}{2}}_{\text{exposure to } x_{ut}} \cdot (\xi_i - \xi_u)$$

High exposure to z_t and x_{ut} make a currency risky: appreciates in good times

- high $z_t \longrightarrow$ high tradable consumption $\longrightarrow$ low marginal utility
- high $x_{ut} \longrightarrow$ high non-tradable consumption in the US $\longrightarrow$ low marginal utility <u>iff $\rho > 1$</u>
- shocks to γ_t purely redistributive, cancel out if $\eta = 1/2$

exchange rates and capital flows

Exchange rate determined by endowments, inflows $\Delta l_{i,t+1}$, and reserve accumulation Δm_{t+1}

$$q_{it} = \alpha e_i z_t - \alpha x_{it} + \underbrace{\alpha L_i \Delta l_{i,t+1}}_{\text{inflows}} - \underbrace{\alpha M_i \Delta m_{i,t+1}}_{\text{reserves}}$$

Reserve accumulation $\Delta m_{i,t+1} \equiv \Delta m_{t+1}$:

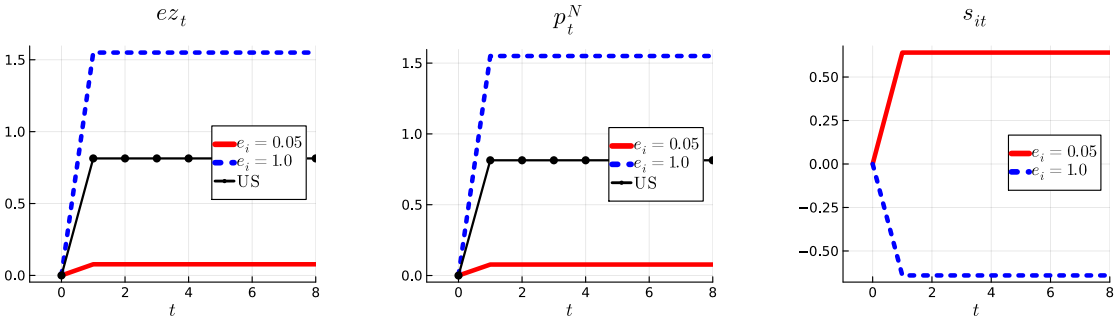
$$\Delta m_{t+1} = \tau(\underbrace{q_t - q_{ut}}_{\text{USD}\downarrow})$$

Capital inflows:

$$L_i l_{i,t+1} = \underbrace{(L_i - L)\gamma_t}_{\text{risk aversion}} + \underbrace{L\Delta m_{t+1}}_{\text{liabilities}} + \underbrace{\chi \mathbb{E}_t[\Delta q_{i,t+1} - \Delta q_{u,t+1} + r_{i,t+1} - r_{u,t+1}]}_{\text{expected excess returns}}$$

Euler equations

commodity shock z_t



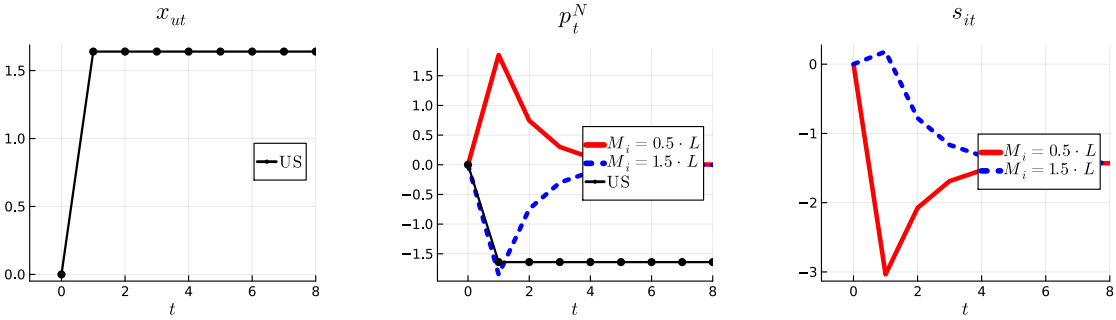
(a) tradable output

(b) price of non-tradables

(c) exchange rate

Figure: Responses of tradable output, relative prices of non-tradable goods, and bilateral exchange rates against USD to a standard innovation to z_t at $t = 1$. The vertical axis in percentage points.

US non-traded output shock x_{ut}



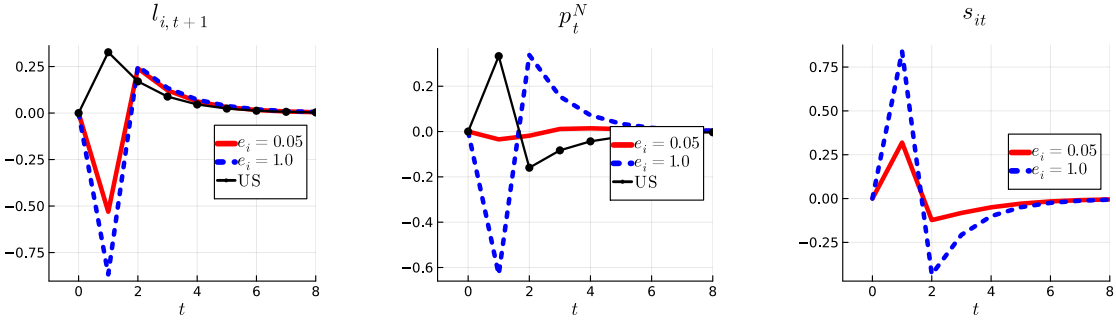
(a) US nontradable output

(b) price of non-tradables

(c) exchange rate

Figure: Responses of non-tradable output, relative prices of non-tradable goods, and bilateral exchange rates against USD to a standard innovation to x_{ut} at $t = 1$. The vertical axis in percentage points.

risk-aversion shock γ_t: by commodity intensity



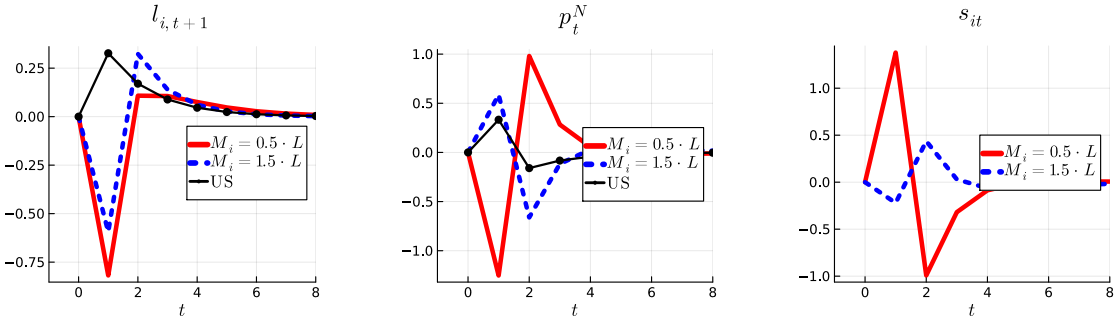
(a) external debt

(b) price of non-tradables

(c) exchange rate

Figure: Responses of financial inflows, relative prices of non-tradable goods, and bilateral exchange rates against USD to a standard innovation to γ_t at $t = 1$. The vertical axis in percentage points.

risk-aversion shock γ_t: by dollar assets



(a) external debt

(b) price of non-tradables

(c) exchange rate

Figure: Responses of financial inflows, relative prices of non-tradable goods, and bilateral exchange rates against USD to a standard innovation to γ_t at $t = 1$. The vertical axis in percentage points.

Factor structure

dollar and commodity factors

Define $\Delta s_{i,t+1} \equiv \Delta q_{i,t+1} - \Delta q_{u,t+1}$ and define the dollar and commodity factors as

$$\text{doll}_{t+1} \equiv \int_{\underline{e}}^{\overline{e}} \Delta s_{i,t+1} di$$

$$\text{comm}_{t+1} \equiv \int_{e>\text{med}\{e\}} \Delta s_{i,t+1} di - \int_{e<\text{med}\{e\}} \Delta s_{i,t+1} di$$

Run the following factor regressions:

$$\Delta s_{i,t+1} = \beta_{i,\text{doll}} \text{doll}_{t+1} + \beta_{i,\text{comm}} \text{comm}_{t+1} + \epsilon_{i,t+1}$$

simplifying assumptions

Let (e_i, M_i) be independent in cross-section, set $\rho_x = \rho_z = 1$ and $\rho_\gamma = 0$

- let e_L and e_H be averages over $[\text{med}\{e\}, \bar{e}]$ and $[\underline{e}, \text{med}\{e\}]$

Consider the limit $\chi \longrightarrow \infty$ with $\chi^{-\frac{1}{2}} M_i \longrightarrow m_i$

- leads to $\chi^{-1} L_i \longrightarrow l_i$ for all i, including u
- endogenous states drop out, analytical solutions for all statistics

Lemma 1

Denote $w_t' = (z_t \ \gamma_t \ x_{ut})$. In equilibrium,

$$q_{it} = \Psi(e_i, m_i)' w_t - \alpha x_{it} + O(\chi^{-\frac{1}{2}})$$

$$q_{ut} = \Phi' w_t + O(\chi^{-\frac{1}{2}})$$

factor construction

Proposition 2

The dollar and commodity factors are

$$\text{doll}_{t+1} = -\alpha \cdot \Delta x_{u,t+1} + \frac{\Gamma \sigma_x^2 \phi^2 (\rho - 1)}{2} \cdot \Delta \gamma_{t+1}$$

$$\text{comm}_{t+1} = -(e_H - e_L)\alpha \cdot \Delta z_{t+1} + \Gamma \sigma_z^2 \phi^2 e (e_H - e_L) \cdot \Delta \gamma_{t+1}$$

Here $\phi = \sqrt{\alpha^2 / (\alpha + (\rho - 1)\alpha)}$.

- dollar factor increases in $\Delta\gamma_{t+1}$: US is "safe" with respect to dollar risk
- commodity factor increases in $\Delta\gamma_{t+1}$: high-commodity countries are output-risky

factor regressions

Proposition 3

Bilateral depreciations are

$$\Delta s_{i,t+1} = \frac{e_i - e}{e_H - e_L} \cdot \text{comm}_{t+1} + (1 - \phi\tau(m_i - l)) \cdot \text{doll}_{t+1} - \alpha\sigma_x\epsilon_{i,t+1}$$

- commodity loadings map to commodity exposure
- dollar loadings map to dollar assets

factor comovement

Proposition 4

The second moments of the factors are

$$\mathbb{V}[\text{doll}_t] = \alpha^2\sigma_x^2 + \phi^4(\rho-1)^2\sigma_x^4/2 \cdot (\Gamma\sigma_\gamma)^2$$

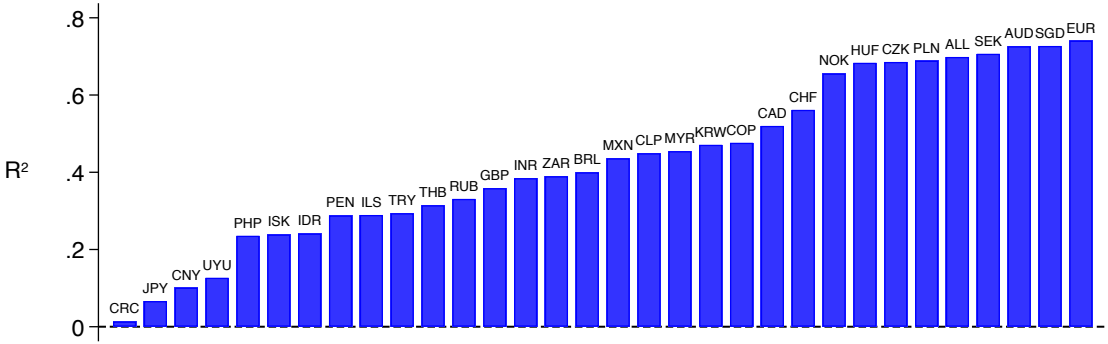
$$\mathbb{V}[\text{comm}_t] = (e_H - e_L)^2\alpha^2\sigma_z^2 + 2\phi^4 e^2(e_H - e_L)^2\sigma_z^4 \cdot (\Gamma\sigma_\gamma)^2$$

$$\mathbb{C}[\text{comm}_t, \text{doll}_t] = \phi^4 e(e_H - e_L)(\rho-1)\sigma_z^2\sigma_x^2 \cdot (\Gamma\sigma_\gamma)^2$$

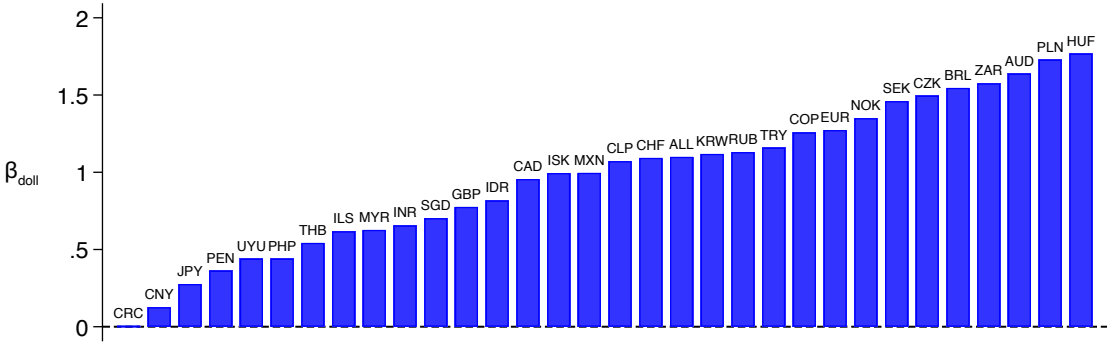
- the map $(\sigma_x^2, \sigma_z^2, (\Gamma\sigma_\gamma)^2) \mapsto (\mathbb{V}[\text{doll}_t], \mathbb{V}[\text{comm}_t], \mathbb{C}[\text{comm}_t, \text{doll}_t])$ is invertible
- factors are only correlated due to $\gamma_t \longrightarrow$ correlation informative on its strength
- variance informative on the strength of shocks to z_t and x_{ut}

Empirical analysis

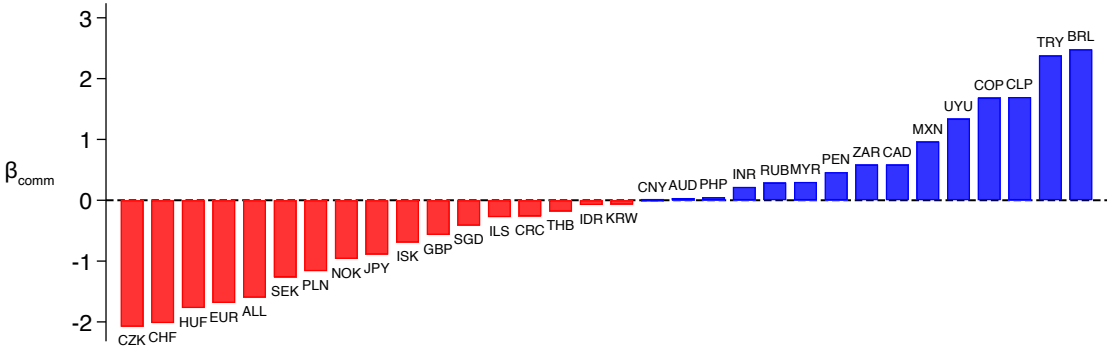
explanatory power



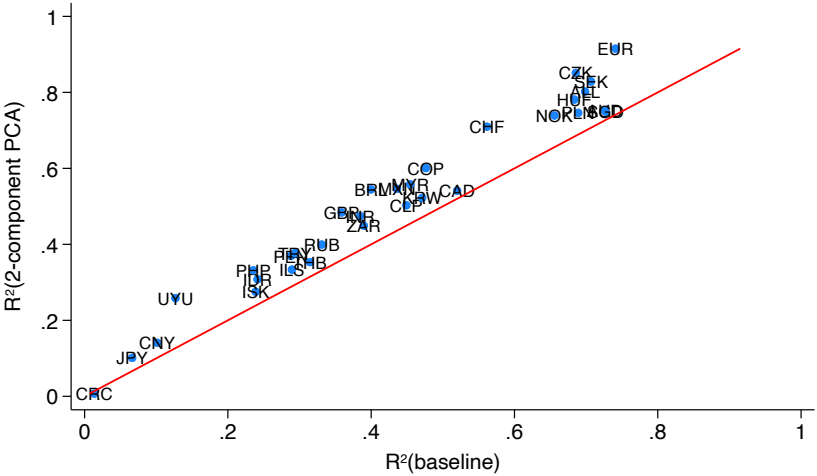
dollar loadings



commodity loadings



benchmarking explanatory power



determinants of loadings

Estimate the following specification for each factor $f = \{\text{comm}, \text{doll}\}$:

$$\beta_{i,f} = \alpha_f + \theta_{f,u}\text{USD}_i + \theta_{f,r}\text{EXP}_i + \epsilon_{i,f},$$

- USD_i is a measure of dollar assets accumulation in country i
- choose share of dollar portfolio-assets in total portfolio-assets for baseline
- EXP_i is a measure of commodity instensity in country i
- choose share of commodity exports in total exports for baseline

commodity factor loadings

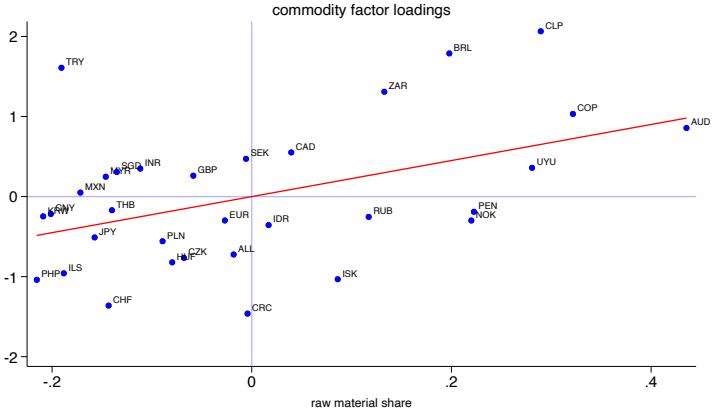


Figure: commodity factor loadings and share of raw materials in exports

dollar factor loadings

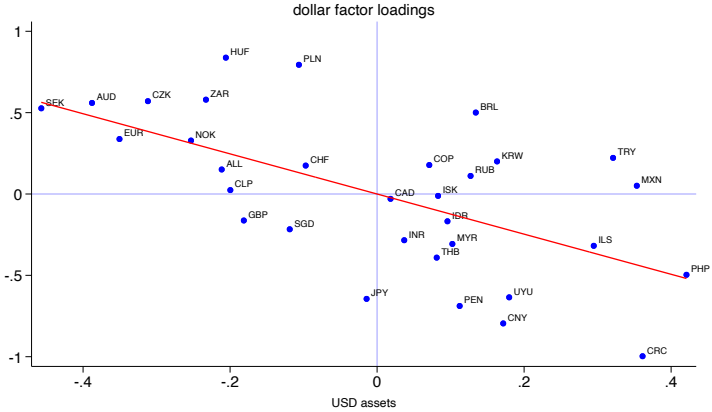


Figure: dollar factor loadings and share USD in foreign assets

commodity loadings

	(1)	(2)	(3)	(4)	(5)	(6)
Commodity factor loading β_{comm}						
USD	2.64***	2.87***	2.85***	2.54***	2.56***	2.51***
	(0.68)	(0.72)	(0.72)	(0.72)	(0.72)	(0.87)
EXP	2.25***	2.80***	2.75***	2.41**	2.40**	2.67**
	(0.79)	(0.89)	(0.93)	(0.87)	(0.93)	(1.21)
R^2	0.56	0.57	0.57	0.65	0.66	0.68
Controls		(1) + cent.	(2) + open.	(3) + size	(4) + dist.	(5) + grav.
Obs.	32	32	32	32	32	32

dollar loadings

	(1)	(2)	(3)	(4)	(5)	(6)
Dollar factor loading β_{doll}						
USD	-1.23^{***}	-1.27^{***}	-1.30^{***}	-1.28^{***}	-1.29^{***}	-1.30^{***}
	(0.27)	(0.29)	(0.28)	(0.29)	(0.29)	(0.31)
EXP	0.81^{**}	0.73^{*}	0.67	0.69	0.69	1.00^{*}
	(0.37)	(0.41)	(0.43)	(0.46)	(0.46)	(0.50)
R^2	0.38	0.38	0.39	0.39	0.40	0.47
Controls		(1) + cent.	(2) + open.	(3) + size	(4) + dist.	(5) + grav.
Obs.	32	32	32	32	32	32

questions

Is this about choosing USD at the base currency?

- EUR and GBP bases

How is this related to the carry-dollar regressions in the literature?

- loadings
- explanatory power

Is commodity intensity related to trade centrality?

- loadings
- explanatory power

Is commodity intensity related to tradable output sensitivity?

- loadings
- explanatory power

Quantitative analysis

calibration

Use the three moments of factors + risk-sharing + excess volatility of exchange rates

Moment	Data	Model
$\text{std}(\text{doll}_t)$	2.27pp	2.27pp
$\text{std}(\text{comm}_t)$	1.43pp	1.43pp
$\text{corr}(\text{doll}_t, \text{comm}_t)$	0.51	0.51
$\text{corr}(\Delta s_{it}, \Delta c_{it} - \Delta c_{ut})$	-0.20	-0.20
$\text{std}(\Delta s_{it})/\text{std}(\Delta c_{it})$	4.86	4.86

risk-sharing moment

parameters

Parameter	Value	Role	Origin
α	0.86	share of non-tradables	$\text{std}(\Delta s_{it})/\text{std}(\Delta c_{it})$
L	6.79	total external assets	$\mathbb{C}[\Delta s_{it}, \Delta c_{it} - \Delta c_{ut}]$
σ_γ	1.42	standard deviation of innovations to γ_t	$\mathbb{C}[\text{comm}_t, \text{doll}_t]$
σ_x	1.64	standard deviation of innovations to x_{it}	$\mathbb{V}[\text{doll}_t]$
σ_z	1.55	standard deviation of innovations to z_t	$\mathbb{V}[\text{comm}_t]$
τ	0.83	stabilization elasticity	reserves
ρ_z	1	persistence of z_t	tradable output
ρ_x	1	persistence of x_{it}	non-tradable output
$\{e_i, M_i\}$		distribution of country types	exports + external assets
ρ	2	inverse of EIS	
β	0.98	discount factor	
ρ_γ	0	persistence of γ_t	

variance decomposition of US price level and factors

Table: variance shares

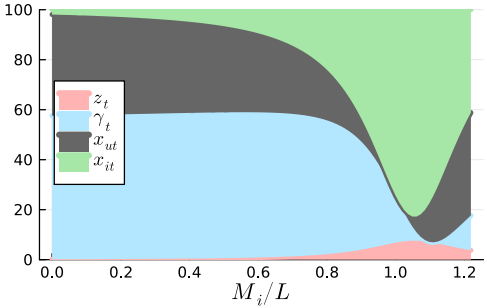
	Δq_{ut}	doll_t	comm_t
$\Delta\gamma_t$	21%	61%	43%
Δz_t	8%		57%
Δx_{ut}	71%	39%	

drivers of local prices

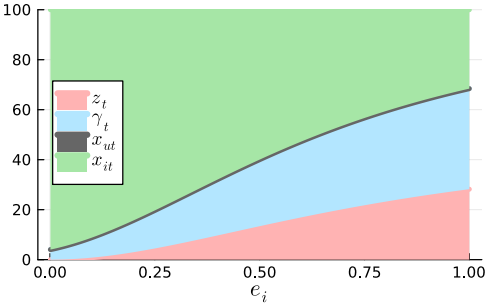
Table: variance decomposition of Δq_{it}

	Low M_i		High M_i	
	Low e_i	High e_i	Low e_i	High e_i
$\Delta\gamma_t$	55%	58%	38%	27%
Δz_t	0%	0%	0%	2%
Δx_{ut}	42%	40%	44%	50%
Δx_{it}	2%	2%	18%	21%
std	9.3%	9.7%	3.3%	3.1%

drivers of local prices in cross-section



(a) shares of non-traded price variance, $e_i = e$



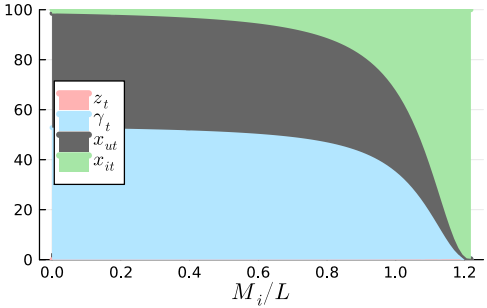
(b) shares of non-traded price variance, $M_i = L$

drivers of bilateral exchange rates

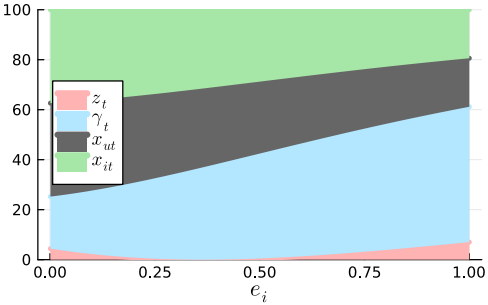
Table: variance decomposition of Δs_{it}

	Low M_i		High M_i	
	Low e_i	High e_i	Low e_i	High e_i
$\Delta\gamma_t$	50%	53%	37%	22%
Δz_t	0%	0%	4%	0%
Δx_{ut}	48%	45%	14%	19%
Δx_{it}	2%	2%	45%	60%
std	10.9%	11.1%	2.1%	1.8%

drivers of exchange rates in cross-section

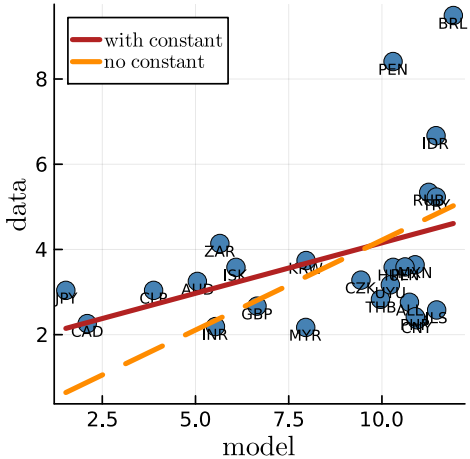


(a) shares of exchange rate variance, $e_i = e$



(b) shares of exchange rate variance, $M_i = L$

explaining volatility in the data



model	0.24**	0.42***
	(0.12)	(0.04)
const	1.79*	—
	(1.05)	
R^2	0.15	0.82
N	25	25

Table: monthly volatility of exchange rates over 1990-2020: data on the model counterpart

conclusion

Macroeconomic model:

- output + financial shocks $\mapsto$ dollar and commodity factors
- heterogeneity in exposure to commodities and dollar assets $\mapsto$ loadings on factors

Replicate factors and loadings in the model

Decompose variation in exchange rates into fundamental sources

- financial shocks consistently important
- financial shocks more important for bilateral exchange rates against USD
- high dollar assets insulate currencies from global shocks

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Euler equations and Backus-Smith

Consumption and exchange rate are related:

$$c_{it} = \frac{1-\alpha}{\alpha} q_{it} + x_{it}$$

Backus-Smith covariance:

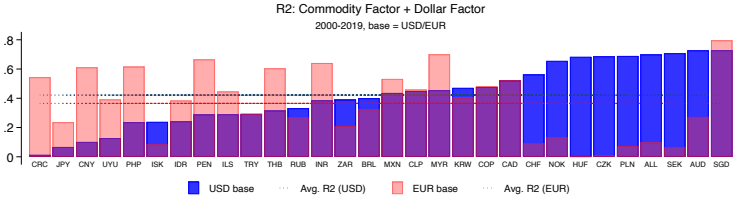
$$\mathbb{C}[c_{it} - c_{ut}, q_{it} - q_{ut}] = \frac{1-\alpha}{\alpha} \mathbb{V}[q_{it} - q_{ut}] + \mathbb{C}[q_{it} - q_{ut}, x_{it} - x_{ut}]$$

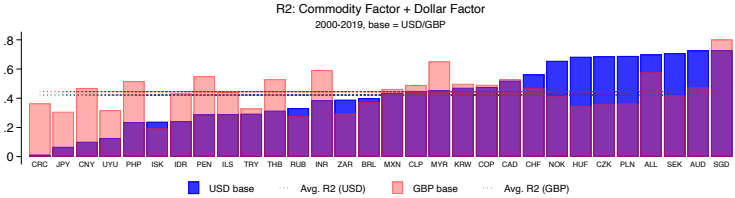
Euler equation leads to

$$r_{it} = \frac{1-\alpha}{\alpha} \mathbb{E}_t[\Delta q_{i,t+1}] + \mathbb{E}_t[\Delta x_{i,t+1}]$$

back

explanatory power for EUR/GBP bases





back

carry explanatory power

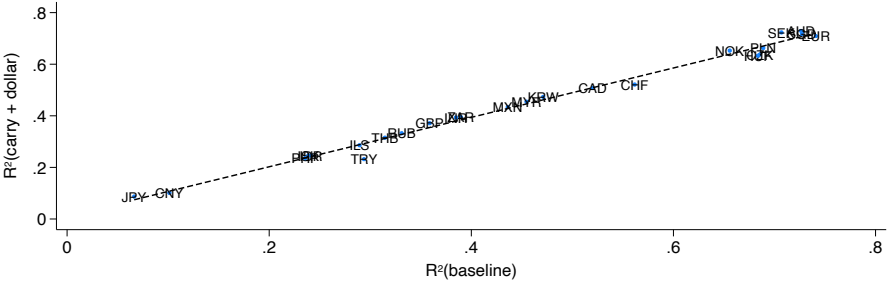


Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the carry factor

back

carry loadings

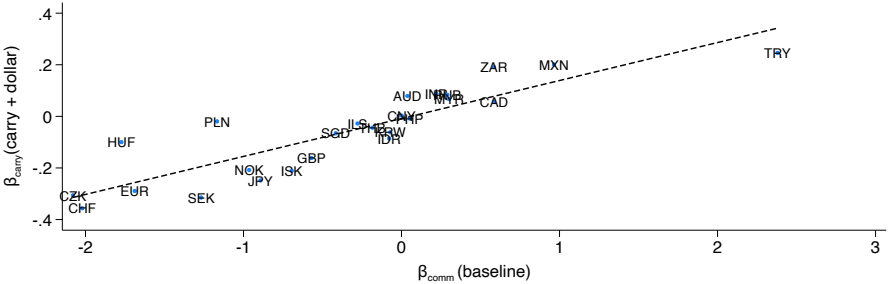


Figure: loadings on the carry factor vs commodity factor, bivariate regressions that include the dollar factor

back

trade centrality explanatory power

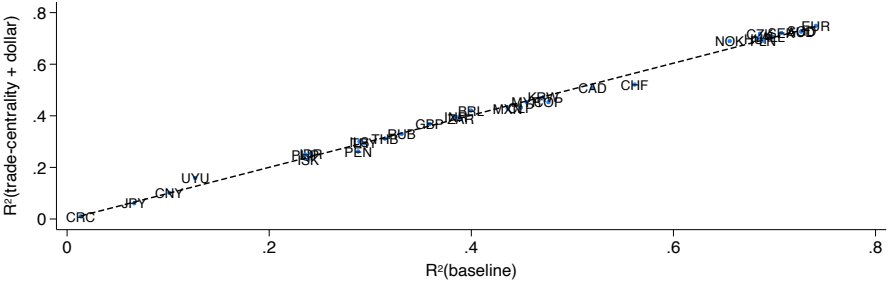


Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the trade centrality factor

back

trade centrality loadings

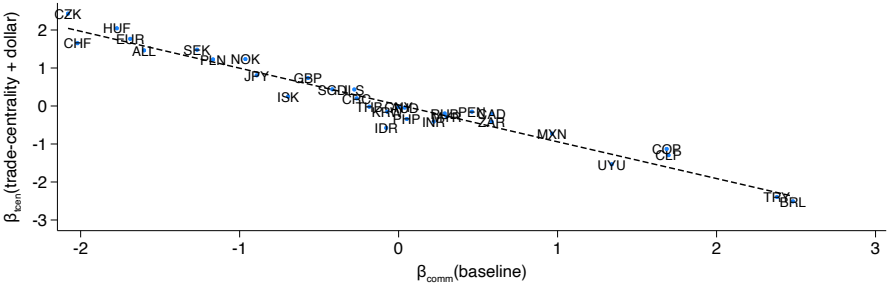


Figure: loadings on the trade centrality factor vs commodity factor, bivariate regressions that include the dollar factor

back

tradable output sensitivity explanatory power

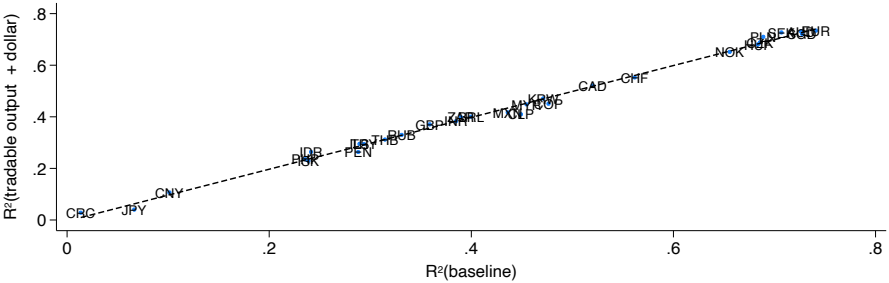


Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the tradable output sensitivity factor

back

tradable output sensitivity loadings

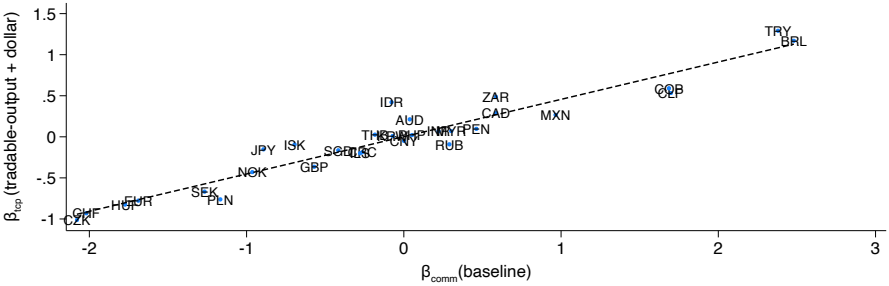


Figure: loadings on the tradable output sensitivity factor vs commodity factor, bivariate regressions that include the dollar factor

back

risk-sharing moment

back

Risk sharing typically assessed by $\text{corr}[\Delta c_{i,t+1} - \Delta c_{u,t+1}, \Delta s_{i,t+1}]$

- equal to 1 with complete and integrated markets
- close to zero (negative) in the data

Proposition 5

The risk-sharing covariance $C_i = \mathbb{C}[\Delta c_{i,t+1} - \Delta c_{u,t+1}, \Delta s_{i,t+1}]$ is

$$C_i = F_x(m_i)\sigma_x^2 - F_z(e_i)\sigma_z^2 - F_\gamma(e_i, m_i)\sigma_\gamma^2$$

Here $F_z(\cdot) > 0$ and $F_\gamma(\cdot) > 0$.

- without shocks to x_{ut}, correlation always equal to -1